



Assessing acid performance in NiMH battery recycling: A comparative study of inorganic and organic acids

Daniel Sánchez Piloto^a, Denise Croce Romano Espinosa^a,
Amilton Barbosa Botelho Junior^{b,c,*}

^a Department of Chemical Engineering, Polytechnic School, University of Sao Paulo, Brazil

^b Department of Materials Science and Engineering, Massachusetts Institute of Technology, United States

^c Department of Chemical Engineering, Norwegian University of Science and Technology, Trondheim, Norway

ARTICLE INFO

Keywords:

Circular economy
Rare earth elements, urban mining
Sustainable mining

ABSTRACT

The growing demand for critical metals has intensified the need for efficient recycling routes for spent NiMH batteries. This study presents a comparative assessment of inorganic (sulfuric acid – H₂SO₄ and phosphoric acid – H₃PO₄) and organic acids (citric acid – C₆H₈O₇ and aminoacetic acid – glycine – C₂H₅NO₂) for the leaching of the black mass obtained after the physical processing of NiMH battery modules. The objective was to identify the most suitable leaching agent considering metal recovery efficiency, reaction parameters, and reagent cost. The novelty of this work lies in the systematic comparison between organic and inorganic acids, including the first-time reported use of glycine for this purpose. Experiments with H₂SO₄ achieved leaching efficiencies of 100 % for rare earth elements (REEs), 97.7 % for Ni, and 99.2 % for Co at 2.0 mol/L, 25 °C, solid-liquid ratio 1/7.5 (g of solid material per L of acid) for 60 min. The use of H₃PO₄ enabled the selective leaching of Ni and Co, while REEs precipitated as insoluble phosphates. Leaching with organic acids proved feasible but demanded the highest temperature (90 °C) and reaction time (120 min) than inorganics to promote protonation and ensure metal solubilization. C₆H₈O₇ had leaching efficiencies of 92.7 % for Ni and 96.7 % for Co, while C₂H₅NO₂ achieved leaching percentages above 98 % in acidic conditions (pH 2.0) and under the lowest solid-liquid ratio (1/20). A preliminary cost analysis indicated that H₂SO₄ provides the most cost-effective leaching among the evaluated acids.

1. Introduction

Given the growing dependence on critical metals for new technologies, the recovery of elements such as Ni, Co, and rare earth elements (REEs) from secondary sources has become essential to reduce supply risks and promote the circular economy. In this context, the recycling of NiMH batteries represents a key opportunity to reclaim these valuable materials and minimize the environmental impacts associated with their primary extraction [1,2].

NiMH batteries had been developed in the 1970s for electronic equipment and were further used in hybrid electric vehicles (HEV) due to their proven reliability, safety, and cost-effectiveness [3,4]. In 2023 (Figure S1, in Supporting Information), the market was valued between \$2.4 and \$3.6 billion and is expected to grow to approximately \$9.0 billion by 2030 [3,5,6].

NiMH batteries are composed of external structure (typically made of

steel or plastic, provides structural integrity and mechanical protection), electrolyte (usually KOH or NaOH that allows the current transfer between the electrodes), separator (usually composed of a blend of polypropylene and polyethylene fibers, or polyamides, ensures electrical insulation between the electrodes, thereby minimizing the risk of short circuits), and the electrodes (anode and cathode that serve as the active sites where electrochemical reactions take place during charge and discharge cycles) [7,8]. Details about the electrodes are depicted in the Supporting Information.

The main approaches for recycling spent NiMH batteries are pyrometallurgical and hydrometallurgical routes. Pyrometallurgy involves operating at temperatures above 1500 °C and yields two main products: a slag containing REEs and a transition metal alloy primarily composed of Fe, Ni, and Co. Both products must subsequently undergo hydrometallurgical treatment to obtain higher-purity materials. The main drawbacks of this method are its high CO₂ emissions and the low purity

* Correspondence to: Department of Chemical Engineering Engineering, Norwegian University of Science and Technology, Norway.

E-mail address: amilton.b.b.junior@ntnu.no (A.B. Botelho Junior).

<https://doi.org/10.1016/j.jece.2026.121066>

Received 31 August 2025; Received in revised form 31 December 2025; Accepted 3 January 2026

Available online 6 January 2026

2213-3437/© 2026 The Authors. Published by Elsevier Ltd. This is an open access article under the CC BY license (<http://creativecommons.org/licenses/by/4.0/>).

of the products. In contrast, hydrometallurgical routes are generally considered more environmentally friendly as they produce lower emissions. These processes typically involve mechanical pretreatment, leaching, and purification stages, enabling efficient and selective recovery of metals with higher purity [9].

In hydrometallurgical battery recycling routes, mechanical pretreatments and physical separation techniques before leaching (e.g., shredding, milling, sieving, magnetic separation) reduces the volume of material to be processed and concentrates the valuable fractions (black mass) rich in critical metals, while also generating separate streams rich in external structures (steel or plastic) and polymers (separators) for recycling by well-established processes [10]. Leaching of NiMH batteries predominantly reports the use of inorganic acids such as H_2SO_4 [2,11], HCl [12,13] and H_3PO_4 [14,15], with H_2SO_4 being the most widely used.

The use of HCl generally requires harsher conditions (concentrations up to 12.0 mol/L and temperatures up to 90 °C) to achieve leaching efficiencies above 95 % for REEs and Ni [10]. In the case of H_3PO_4 , it is used primarily to selectively leach transition metals (Ni, Co, Mn, Fe, and Zn) and Al from NiMH electrodes, as REEs precipitate in the presence of PO_4^{3-} ions forming $\text{REEPO}_4 \cdot x\text{H}_2\text{O}$ [16], but scarcely discussed in the literature and presents an interesting alternative due to its ability to separate REEs from other metals in the black mass.

Organic acids have also been investigated, including formic acid (CH_2O_2) [17], citric acid ($\text{C}_6\text{H}_8\text{O}_7$) [18], acetic acid ($\text{C}_2\text{H}_4\text{O}_2$) [19] and oxalic acid ($\text{C}_2\text{H}_2\text{O}_4$) [20], due to their lower environmental impact if produced from natural sources or biotechnological processes making them promising alternatives to reduce their carbon footprint [21]. Additionally, they are generally less corrosive, reducing equipment maintenance costs and extending the operational lifetime of processing infrastructure [22].

Although most studies report lower leaching efficiencies for organic acids compared to inorganic acids, there is still limited research exploring how process parameters—such as pH, temperature, solid-to-liquid ratio, time, and acid concentration—affect their performance. A systematic comparison of these acids under varying operational parameters remains limited. Our study addresses these gaps by establishing a comparative evaluation of leaching efficiency between selected inorganic (H_2SO_4 and H_3PO_4) and organic acids ($\text{C}_6\text{H}_8\text{O}_7$ and $\text{C}_2\text{H}_5\text{NO}_2$), aiming to identify the most effective and economically feasible lixiviant for the recovery of Ni, Co, and REEs from NiMH battery black mass. By including glycine, this work introduces a novel organic-acid-based approach, expanding the range of potential sustainable leaching agents and providing new insights into process optimization for NiMH battery recycling. The comparison will be carried out by varying several parameters (concentration, temperature, S/L ratio, and time) to assess their influence on the leaching process. This study meets the goals for sustainable mining [23].

H_2SO_4 was chosen for comparison, as it is the most used for NiMH battery leaching, offering higher efficiency than other inorganic acids (e.g., HCl , H_3PO_4 , HNO_3) [24] and its lower environmental impact [25]. H_3PO_4 was also studied due to the low solubility of REEs in phosphate solution, which can favor the selective leaching of transition metals such as Ni, Mn, and Co [26]. The selection of organic acids ($\text{C}_6\text{H}_8\text{O}_7$ and $\text{C}_2\text{H}_5\text{NO}_2$) was motivated by their biodegradability and potential for sustainable production through biotechnological processes [21], and the limited literature on their application to NiMH battery leaching, placing sustainable chemical engineering advances in future processes allied to the development of new chemical reaction understanding. Additionally, $\text{C}_6\text{H}_8\text{O}_7$ has shown promising results in Li-ion battery recycling [27–29], suggesting its suitability for the development of organic-acid-based processes capable of treating both NiMH and Li-ion batteries. Such an approach could represent a significant advance toward more versatile and environmentally friendly recycling technologies. Notably, $\text{C}_2\text{H}_5\text{NO}_2$ is proposed for the first time as a leaching agent for NiMH batteries, expanding its application beyond its previously reported use in Li-ion

[30] and NiCd [31] batteries.

The physical processing before leaching was the same as that developed by our group for Li-ion battery recycling, aiming to make a flexible recycling process (for batteries of electric and hybrid vehicles). This study not only evaluates leaching efficiency but also considers reagent cost. The results are expected to provide comparative data that supports the selection of the most technically and economically feasible lixiviant for the hydrometallurgical recovery of REEs, Ni, and Co, and open new research opportunities for the use of organic acid-based leaching systems in NiMH battery recycling. Environmental impact and complete technical economic analysis are beyond our scope in this study, despite its relevance, particularly when comparing inorganic and organic acids, and it should be better compared as an entire process (including separation/purification and product obtaining) instead of only the leaching stage.

2. Materials and methods

A NiMH battery from a 2013 Toyota Prius HEV (model CBU) was received for this study and the flowchart of the developed recycling process is shown in Fig. 1. The battery pack was discharged, and the internal components were separated for characterization. Three discharged modules were ground, and physical separation was performed to obtain a black mass rich in Ni, Co and REEs for leaching experiments.

2.1. Characterization

The external structure and the modules of the battery pack (37.2 kg) were manually separated (Figure S2 in Supporting Information). The module set (28 modules) was discharged over 24 h using four resistors connected in series to prevent overheating and reduce the risk of explosion during discharge, as developed by our group [32–34]. This type of battery has a nominal voltage of 201.6 V, and each module corresponds to 7.2 V [35]. It was measured with a multimeter showing 151 V at the start of the process and less than 1 V by the end. After discharge, the modules were manually separated. One of the modules was divided into six cells, and its internal components were extracted. The cathodes, anodes, internal plastics, and separator sheets were dried for 48 h in a fume hood to allow the evaporation of the electrolyte (Figure S3).

X-ray diffraction (XRD, Rigaku Miniflex 300) analysis was performed to identify the crystalline phases of the anode and cathode active materials at 2 °/min, 15–80° (2 θ), in continuous mode, at 30 kV and 20 mA. For the XRD analysis, the anode material was separated from the support plate manually, while the cathode was ground in an agate mill, resulting in a mixture of active material and support plate. Scanning electron microscopy coupled with energy dispersive spectroscopy (SEM/EDS, ProX Phenom) was carried out after ethanol washing and drying at 60 °C for 24 h to identify the elements present in the battery electrodes.

All the anodes (13) and cathodes (12) from a cell were digested in aqua regia (1 HNO_3 /3 HCl , v/v) at room temperature with a solid/liquid (S/L) ratio of 1/20 for 24 h. The electrode digestion for characterization was performed in triplicate. After digestion, the solution was filtered and diluted in 3 % v/v HNO_3 for chemical analysis in ICP-OES (Agilent Technologies, 710) to quantify La (379.082 nm), Ce (418.659 nm), Nd (406.108 nm), Pr (410.072 nm), Y (371.029 nm), Al (396.152 nm), Co (228.615 nm), Fe (238.204 nm), K (766.491 nm), Mn (259.372 nm), Ni (231.604 nm), and Zn (213.857 nm). The digestion residues were further analyzed by SEM/EDS.

2.2. Milling and physical processing

The milling and physical processing stage aimed to separate the battery components according to their physical properties into different streams, concentrating the active material in a specific product stream. This stage was based on the process developed for Li-ion batteries by Romero et al. [34], adapting a well-established process developed by our

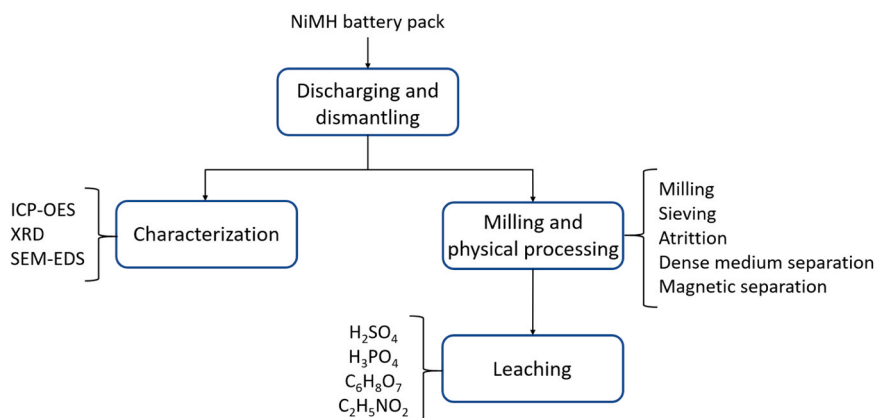


Fig. 1. Flowchart of the recycling route developed in this work.

group, demonstrating its applicability to battery recycling of both electric and hybrid vehicles. The flowchart of the milling and physical processing stage is shown in the [Supporting Information \(Figure S4\)](#).

Milling (3.1 kg) of discharged modules was performed using a Rone FA 2305 knife mill (220 V, 400 W) equipped with a 9 mm mesh screen. The milled material was collected in two containers, corresponding to particle sizes smaller and larger than 9 mm. The physical processing began with the attrition step for 8 min to promote the disaggregation of agglomerates formed during milling. Further, a dense medium separation was carried out in a flotation cell, yielding two fractions: floated and sink. Both fractions and the material retained in the knife mill were washed and sieved (600 μm) to isolate the active material from other components. The resulting aqueous suspension was decanted, filtered, and the solid was dried in an oven at 60 $^{\circ}\text{C}$ for 24 h. The physical process concluded with a magnetic separation step (1500 G) applied to separate metallic Fe and Ni from the plastic components of the battery.

All solid products obtained from the physical processing, as well as 1 L of water used in the decantation and filtration stage, were characterized by ICP-OES after acid digestion to determine their chemical composition (chapter 2.1). The stream enriched in active material (referred to as black mass) from both battery electrodes was selected as the feed for the leaching experiments. This stream was analyzed by XRD applying the Rietveld refinement method to identify and quantify the phases.

To quantify the soluble material in the NiMH battery modules, another module was milled, and the resulting material was washed with 20 L of water for 30 min. After washing, the material was dried in an oven at 60 $^{\circ}\text{C}$ for 24 h, and the wash water was analyzed by ICP-OES to determine the concentrations of La, Ce, Nd, Pr, Y, Al, Co, Fe, K, Mn, Ni, and Zn.

2.3. Leaching experiments

Leaching experiments were conducted in a 100 mL three-necked and round-bottomed reactor with constant magnetic stirring. A condenser was attached to prevent the loss of water/acid by evaporation, and a thermometer was used to monitor the temperature. Experiments were carried out with 5 g of the stream M2 ([Figure S4](#)), and the use of inorganic (H_2SO_4 and H_3PO_4) and organic ($\text{C}_6\text{H}_8\text{O}_7$ and $\text{C}_2\text{H}_5\text{NO}_2$) acids as leaching agents was investigated to enable a comparative evaluation of efficiencies. The effect of variations in acid concentration, temperature, S/L ratio, and reaction time was also evaluated ([Table 1](#)). For $\text{C}_2\text{H}_5\text{NO}_2$, the effect of pH was also assessed by adding HNO_3 or NaOH .

After the reaction time, the solution was filtered through a vacuum filtration system, and the pH was measured. The solid phase was washed with ultrapure water and dried for 24 h in a fume hood at room temperature and further digested in aqua regia (S/L ratio 1/10, 25 $^{\circ}\text{C}$, 24 h, and under constant stirring) for chemical analysis. The leachate,

Table 1

Acids and parameters availed in the leaching stage of the M2 stream.

Acid	Concentration (mol/L)	Temperature ($^{\circ}\text{C}$)	S/L	Time (min)	pH modifier
H_2SO_4	1.0	25	1/5	30	-
	2.0	45	1/	60	-
	3.0	65	7.5	90	-
	4.0	90	1/	120	-
			10		-
H_3PO_4	1.0	25	1/5	30	-
	2.0	45	1/	60	-
	3.0	65	7.5	90	-
	4.0	80	1/	120	-
			10		-
$\text{C}_6\text{H}_8\text{O}_7$	1.0	25	1/5	30	-
	2.0	45	1/	60	-
	3.0	65	7.5	90	-
	4.0	90	1/	120	-
			10		-
$\text{C}_2\text{H}_5\text{NO}_2$	2.0	25	1/	120	HNO_3 (pH 2,0)
			10		
		90	1/	20	NaOH (pH 10,0)
			20		

washing water, and digestion of the leaching residue were analyzed by ICP-OES for mass balance. The leaching percentage was calculated as the ratio between the mass of the element in the leachate and the mass of the element in the initial solid ([Eq. 1](#)).

$$\% \text{Leaching} = \frac{m_{\text{leachate}}}{m_{\text{initial solid}}} * 100 \quad (1)$$

3. Results and discussions

3.1. Characterization of the NiMH batteries

The battery pack consisted of 28 modules connected in series, representing 77 % of the total battery mass, with each module weighing approximately 1.03 kg. Each module contained 6 cells, which were separated by a plastic column, and the electrodes were attached to steel side bars. Each cell contained 13 anodes, 12 cathodes, 12 separators, and 2 internal plastics. The separators surrounded the cathodes preventing contamination and contact between the electrodes. The internal plastics held all the parts of the cell immersed in the electrolyte ([Table 2](#)).

The percentages of the external case, anode, and cathode in the cell align with those reported in the literature for the same model of NiMH HEV batteries [36]. It is important to note that Larsson et al. [36] consider the mass of the anode support plate as part of the steel bars, which explains the difference in the data ([Table 2](#)). In the reviewed literature, only one study [36] was found that reported the mass

Table 2

Components presented in the NiMH battery cell determined after dismantling a discharged cell compared to the literature.

Cell component	Mass in the module (g)	Mass in the cell (g)	Mass percentage in the cell (%)
	This study		Larsson et al. [36]
Anode	291.4	48.6	28.5
Cathode	280.8	46.8	27.4
Separator	95.3	15.9	9.3
Internal plastics	6.5	1.1	0.6
External case	182.4	30.4	17.8
Steel bars	115.4	19.2	11.3
Evaporated electrolyte.	51.7	8.6	5.1

percentages of the internal components of a NiMH HEV battery cell, as most authors work directly with the battery's black mass.

LaNi₅ metal alloy was identified as the active material in the anodes of the NiMH battery (Figure S5a), while peaks of Ni(OH)₂ and Ni were identified in the cathode (Figure S5b). Ni(OH)₂ corresponds to the active material, and metallic Ni corresponds to the support plate. These results align with those reported [36–38] for NiMH HEV batteries and electronic devices.

SEM/EDS analysis showed peaks corresponding to Ni, C, La, Ce, Nd, Pr, Y, K, Co, Mn, and Al were identified (Figure S6a). The C present in the anode is from an organic binder additive, which is commonly added to improve cohesion and ensure uniform distribution of the active material in the electrode [36]. K is part of the electrolyte composition. Y₂O₃ is added to enhance corrosion resistance [39]. In the cathode sheet (Figure S5b), Ni was detected as hydroxide (Ni(OH)₂, spherical particles) as the active material. Co was also detected, as it is often used as a partial substitute for Ni to enhance performance and conductivity [7, 40].

The battery cell contains Ni as the main element (44.4 % in the anodes and 64.0 % in the cathodes) while REEs were found in the anode (23.4 %), with La the highest concentrated - 14.6 % (Table 3). Our results are similar to those reported by Korkmaz [35] and Larsson et al. [36]. The main difference between the studies is the Fe content as the anode support plate of our samples, due to the digestion with the active material, whereas Korkmaz [35] and Larsson et al. [36] only digested the active material.

The digestion of the anode was not complete, leaving an undissolved residue corresponding to 2 % of the initial mass. SEM/EDS analysis of this residue (Figure S7) revealed peaks of C and Cl. C corresponds to the organic binder additive used to enhance the cohesion of the anode active material, while Cl is attributed to residual HCl from the aqua regia used

Table 3

Mass percentage of chemical elements in the NiMH battery electrodes in a battery module (28.7 kg).

Element	This study		Larsson et al. [36]		Korkmaz [35]	
	Anode	Cathode	Anode	Cathode	Anode	Cathode
Al	1.5 ± 0.08	-	1.5	0.1	1.8	0.1
Ce	6.2 ± 0.41	-	7.4	-	5.7	-
Co	1.8 ± 0.15	4.4 ± 0.37	3.6	5.7	5.6	5.5
Fe	22.5 ± 0.63	-	0.1	-	0.1	0.1
K	1.2 ± 0.17	1.9 ± 0.12	0.4	0.2	0.1	0.6
La	14.6 ± 0.45	-	20.2	-	19.3	-
Mn	2.8 ± 0.24	-	5.6	0.3	4.7	0.1
Ni	44.4 ± 1.31	64.0 ± 1.91	52.3	64.7	64.3	76.4
Nd	1.6 ± 0.15	-	2.4	-	2.7	-
Pr	0.5 ± 0.07	-	1.0	-	2.7	-
Y	0.6 ± 0.11	0.3 ± 0.09	0.7	0.9	0.8	0.4
Zn	-	0.5 ± 0.08	-	0.5	0.6	3.7

during digestion.

For every 28.7 kg of modules (1 battery pack), there are 8.6 kg of Ni, 2.0 kg of REEs, and 0.5 kg of Co. Based on this composition, the processing of 1 ton of NiMH modules would have an estimated value of \$4513.3 in Ni, \$496.0 in Co, and \$1336.9 in a battery-level mix of REEs. Metal prices were sourced from the Shanghai Metals Market on April 14, 2025 [41]. It shows the economic potential of recycling and decreasing the dependence on mining. In addition, the metal content (REEs and Ni) in spent NiMH batteries is higher than in natural resources [23,42].

3.2. Milling and physical processing

The comminution and physical processing stage (Fig. 2) aimed to separate the active material of the NiMH battery electrodes for hydrometallurgical processing. We adapted the process of Li-battery recycling to create a flexible process for different batteries from vehicles. Our goal was to separate valuable compounds (e.g., rich in Ni, Co, and REEs) from low-valuable materials (e.g., polymer and steel) without degrading materials using thermal steps.

Two products were obtained from knife milling: the material retained on the mesh and the solid comminuted material with a particle size smaller than 9 mm. The losses in the comminution stage were 5.4 %, including evaporated electrolyte and part of the solid material (lost as dust). The retained material (>9 mm, 17.3 %) was composed of the external case and steel bars that were not comminuted. From the comminuted material (77.3 %), three streams were obtained: a separator fraction (stream M3, 2.7 %), a magnetic fraction (stream M4, 21.2 %), and a non-magnetic fraction (stream M5, 1.8 %). The active material recovered from both the comminuted and retained fractions was combined, resulting in a single active material stream (stream M2, 35.3 %). Figure S8 shows the stream products obtained from the milling and physical processing stage (R1, R2, M1, M2, M3, M4, and M5). The distribution of components in the products and the percentage of each element in the streams are presented in Tables S1 and S2.

For hydrometallurgical processing, we separated streams based on their composition. The separator material and the external case were concentrated in the streams R2, M1, M3, and M5. The valuable metals (as REEs, Ni, and Co) were concentrated in the streams M2 and M4. Ni was distributed between the streams M2 (72.3 %) and M4 (25.9 %). Additionally, 97.5 % of the REEs and 83.4 % of the Co were concentrated in the M2 stream. The water used in physical processing did not carry any of the battery's components of interest, which can be recirculated.

At the end of the physical processing stage, a mass loss of 13.7 % was observed. Washing another module, comminuted separately, resulted in a 14.0 % mass loss, a similar value to that obtained for the three previously processed modules. This loss was attributed to the removal of soluble material from the modules. Chemical analysis of the wash water detected only K, indicating that the soluble fraction corresponds to the battery electrolyte (KOH).

The stream R1, consisting of 96.1 % Fe and 3.7 % Ni, could be used for Ni-plated steel production [41]. The stream M4, primarily composed of Fe (61.1 %) and Ni (31.9 %), could be used for Fe-Ni alloys production [41]. Physical processing generated a stream (M2) rich in REEs (18.8 %), Ni (56.4 %), and Co (4.0 %), which was directed to hydrometallurgical treatment for our leaching experiments. The material corresponds to LaNi₅ and Ni(OH)₂ – 43.5 % and 56.5 %, respectively, but not considering the content of Y₂O₃, Fe, and KOH (Figure S9).

Comparing the processing of NiMH battery modules with Li-ion battery cells [34], some differences were observed due to the composition of the modules. In the case of NiMH batteries, the separator is a double sheet that wraps around the cathode and tends to settle at the bottom of the flotation cell along with the cathode metals, requiring an additional dry sieving step (6.30 mm) for its removal. The magnetic separation process was carried out with a lower magnetic field intensity (1500 G) in contrast to Li-ion batteries (13,000 G) since Ni and Fe are

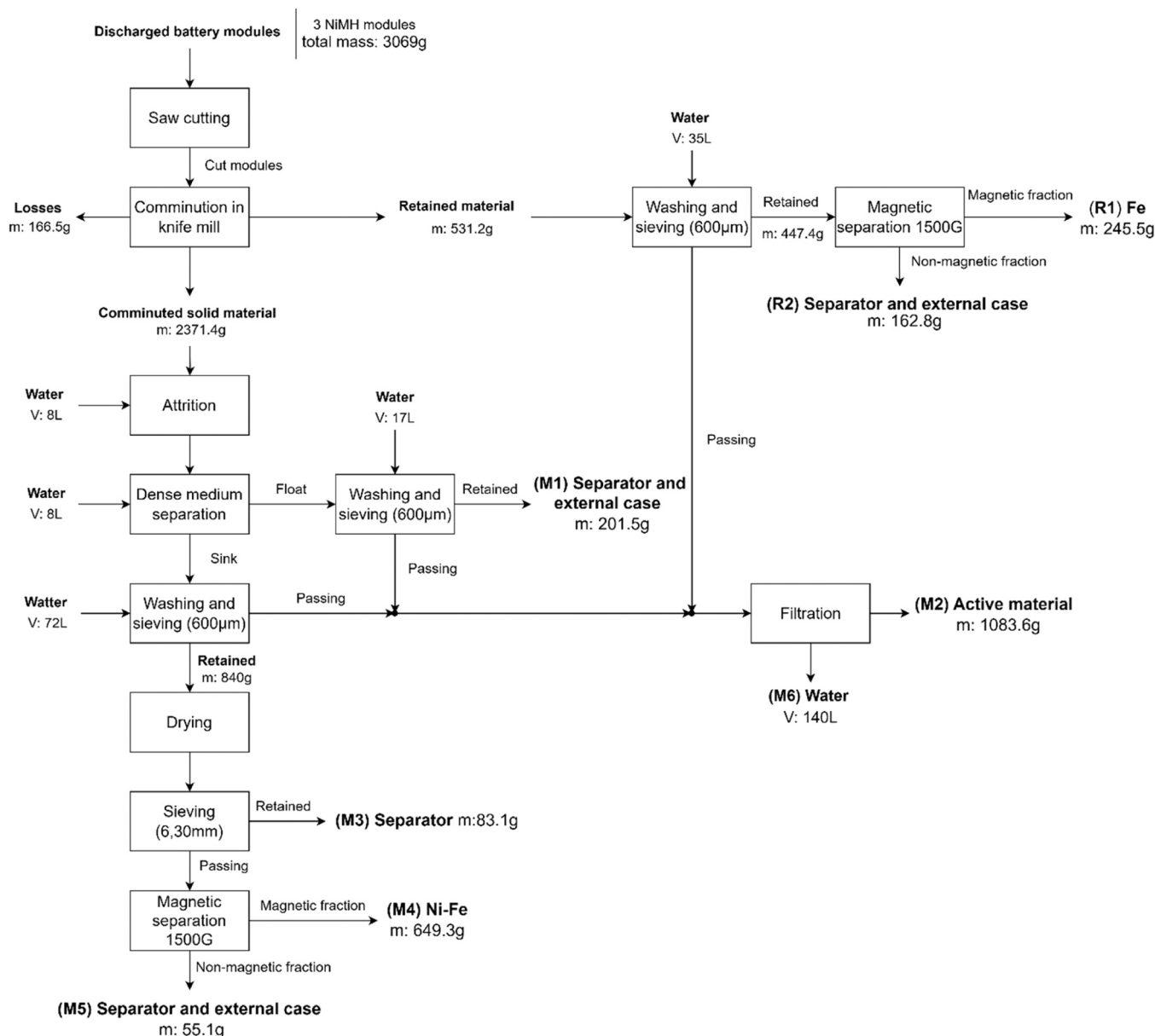


Fig. 2. Results of milling and physical processing stage of NiMH battery.

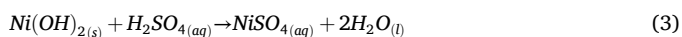
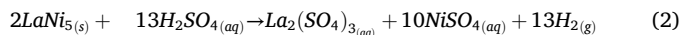
magnetic materials, and the same magnetic field intensity was not needed for their separation.

3.3. Leaching experiments

Different acids (inorganic and organic) were studied, focusing on their key differences and identifying the most technically viable option for future purification studies and potential scale-up. The chemical behavior of the systems was simulated using the Hydra-Medusa software to predict the predominant aqueous species under the experimental conditions for Ni, Co, and La (Figures S10 – S13). La, being the REE present at the highest concentration, was used as a representative of the other REEs. For each acid, the simulations were performed using the concentrations and S/L ratios defined as optimal in this study. Since the software database does not include thermodynamic data at higher temperatures, all simulations were carried out at 25 °C. The final pH measured in each experiment, and the stoichiometric acid ratio (acid used/stoichiometric acid), are presented in Table S3. Experimental error for leaching was 1 % for REEs and 2 % for Ni and Co.

3.3.1. H_2SO_4 leaching

The lowest leaching efficiency for all metals was observed at 1.0 mol/L, reaching 91.0 % – 95.3 % for REEs (La, Ce, Nd, Pr, and Y), 90.7 % for Co, and 65.0 % for Ni (Fig. 3a) because of low stoichiometric ratio (0.8, Eqs. 2–3) and consequently high pH value (4.4).



The highest leaching percentages were achieved at 2.0 mol/L H_2SO_4 (1.6 stoichiometric acid ratio) for all elements reaching 97.8 % – 98.6 % for REEs, 97.8 % for Ni, and 99.7 % for Co. Gasser & Aly [43] reported similar leaching efficiencies for Co (97.6 %), Nd (95.6 %), and La (92.5 %) under the same conditions but observed a lower leaching percentage for Ni (76.6 %) due to the passivation of metallic Ni (forming NiO) present in the cathode support plate [39]. In contrast, we removed the metallic Ni during physical processing via magnetic separation, which has a greater advantage than previous studies.

Increasing the H_2SO_4 concentration reduces the solubility of REEs,

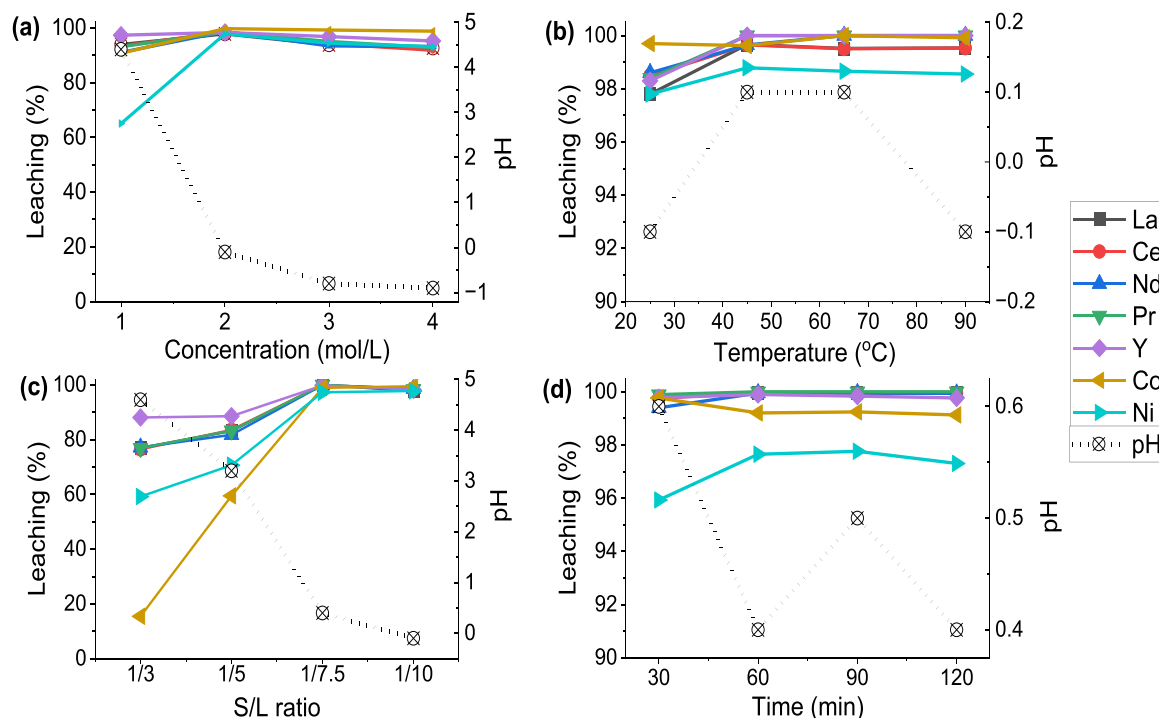
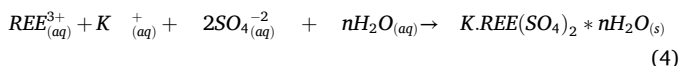


Fig. 3. Leaching efficiency of REEs, Ni and Co with H_2SO_4 by (a) varying concentration (1.0 – 4.0 mol/L), at 25 °C, S/L ratio of 1/10 and 120 min; (b) varying temperature (25 – 90 °C), acid concentration of 2.0 mol/L, S/L ratio of 1/10 and 120 min; (c) varying S/L ratio (1/3–1/10), at 25 °C, acid concentration of 2.0 mol/L for 120 min; (d) varying leaching time (30 – 120 min), at 25 °C, acid concentration of 2.0 mol/L and S/L ratio of 1/7.5.

leading to their precipitation as insoluble sulfate salts [16]. The formation of precipitates was observed in H_2SO_4 concentrations of 3.0 and 4.0 mol/L containing S, Ni, Al, K, La, Ce, Nd, and Pr (Figure S14) as double REE salts in the presence of K^+ and sulfate (SO_4^{2-}) [24,39] as described in Eq. 4.



The leaching efficiencies achieved above 97 % in all temperatures studied (Fig. 3b). Therefore, 25 °C was selected for subsequent experiments, as it minimizes energy consumption. Similar results were reported by Takano et al. [2] who demonstrated that under pH conditions between 0 and 1.5 in the presence of H_2SO_4 , at temperatures from 40 °C to 90 °C, with an S/L ratio of 1/10 for 180 min, the leaching efficiency of NiMH battery electrodes remained stable over 90 % for Ni, 97 % for Co and Y, and approximately 100 % for La and Ce. Sobianowska-Turek [44], also observed consistent leaching efficiencies for Ni, Co, Mn, Zn, La, and Ce using 3.0 mol/L H_2SO_4 , an S/L ratio of 1/10, a reaction time of 75 min, and temperatures of 25 °C and 55 °C.

An increase in metal leaching efficiency (Fig. 3c) and a decrease in pH (Table S3) were observed as the S/L ratio varied from 1/3–1/7.5 due to different stoichiometric acid ratios for each S/L ratio. At S/L ratios of 1/3 and 1/5 and concentration 2.0 mol/L, leaching occurred under sub-stoichiometric conditions (0.5 and 0.8, respectively – Table S3). Leaching efficiencies above 99 % were achieved for REEs and Co, and over 97 % for Ni at an S/L ratio of 1/7.5 (1.2 stoichiometric acid ratio), which was selected as the optimal condition for leaching the M2 stream.

The reaction time has no influence on the leaching efficiency of the active material stream between 60 min and 120 min, with values exceeding 97 % for all metals (Fig. 3d). Based on the leaching tests, the optimal process conditions were established as follows: 2.0 mol/L H_2SO_4 , at 25 °C, S/L ratio of 1/7.5, and 60 min.

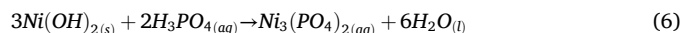
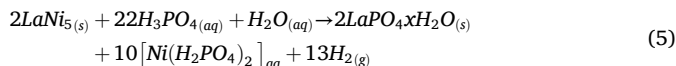
3.3.2. H_3PO_4 leaching

The leaching with H_3PO_4 solution aimed at separating the REEs from

the other metals present in the M2 stream towards the formation of insoluble REE phosphates. Our results demonstrated successful leaching followed by selective precipitation of REEs. In the H_3PO_4 leaching experiments, varying acid concentration, unlike those with H_2SO_4 , the temperature was fixed at 80 °C—identified by Tanji et al. [15,45] as optimal for leaching the black mass of NiMH batteries—with a reaction time of 120 min and S/L ratio of 1/10. Subsequently, the effects of S/L ratio, reaction time, and finally temperature were evaluated (Fig. 4).

Less than 2 % of the REEs were leached while the leaching percentages of Ni and Co increased with higher acid concentrations (Fig. 4a), consistent with the thermodynamic simulations of the system (Figure S11). The formation of insoluble REEPO_4 occurs in two stages: first, the REEs are dissolved upon contact with H^+ ions released by the acid protonation, and then, the released REE ions react with phosphate ions to form insoluble REEPO_4 compounds [14].

The stoichiometric acid ratios (Table S3) were calculated based on the reactions of the LaNi_5 alloy and $\text{Ni}(\text{OH})_2$ with H_3PO_4 (Eqs. 5 – 6). Therefore, the calculation is approximate since it does not consider the amount of acid required for the first stage of leaching the metal alloy.



In the experiment conducted with a concentration of 4.0 mol/L H_3PO_4 , it was possible to leach 98 % of Ni, 99 % of Co, and only 0.1 % of La and 1.5 % of Y, while Ce, Nd, and Pr were not leached, showing a selective leaching of transition metals and selective precipitation of REEs. $\text{LaPO}_4 \cdot 0.5\text{H}_2\text{O}$ was identified in the leaching residue by DRX analysis (Figure S15). According to the chemical analysis of this residue, the main contaminants are Ni, Mn, and polymeric material derived from the undigested separator that was not completely removed during the physical processing stage.

An increase in the leaching percentage of transition metals was observed with the variation of the S/L ratio from 1/5–1/10 (Fig. 4b,

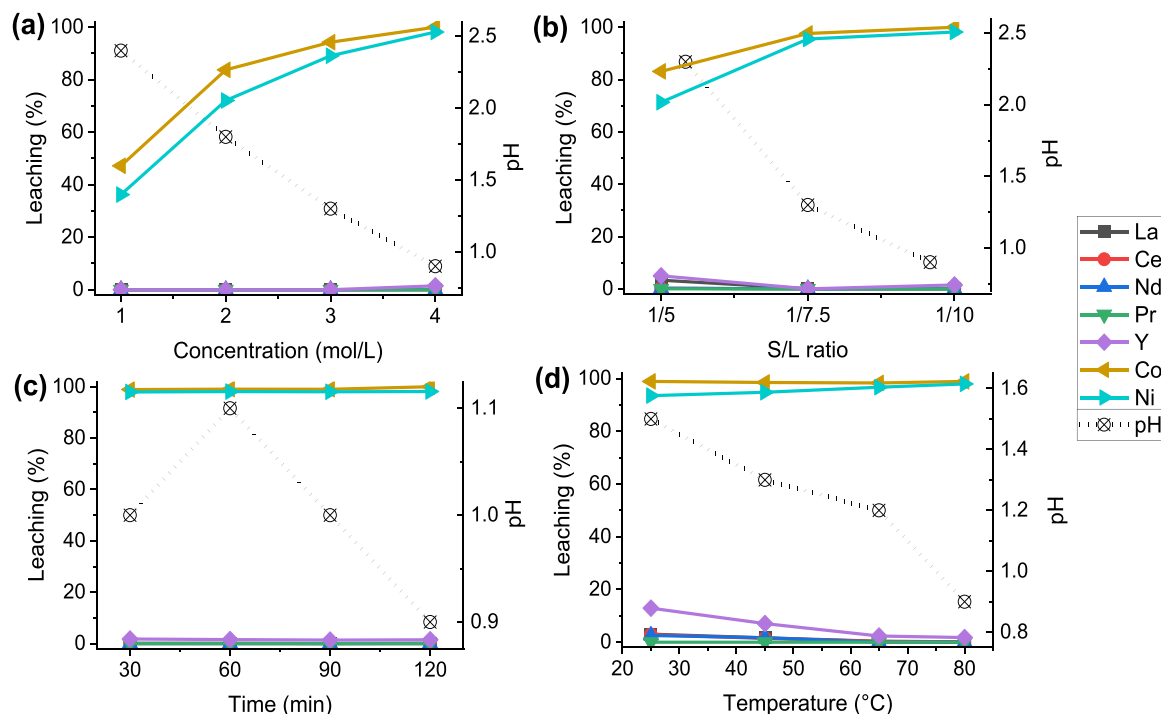


Fig. 4. Leaching efficiency of REEs, Ni and Co with H_3PO_4 by (a) varying concentration (1.0 – 4.0 mol/L), at 80 °C, S/L ratio of 1/10 and 120 min; (b) varying S/L ratio (1/5 – 1/10), at 80 °C, acid concentration of 4.0 mol/L for 120 min; (c) varying leaching time (30 – 120 min), at 80 °C, acid concentration of 4.0 mol/L and S/L ratio of 1/10; (d) varying temperature (25 – 80 °C), acid concentration of 4.0 mol/L, S/L ratio of 1/10 and 30 min.

stoichiometric acid ratio of 1.3, 2.0, and 2.6), a behavior consistent with the results reported by Lie & Liu [14] for an S/L ratio from 1/5–1/20. By using 2.5 mol/L H_3PO_4 , at 25 °C, a 1/10 S/L ratio for 60 min, Lie & Liu [14] achieved leaching efficiencies of 60 % for Ni and Co, lower than that obtained in our study (71 % for Ni and 83 % for Co) due to different acid concentration (4.0 mol/L in our study). A decrease in pH (Table S3) was also observed, varying the S/L ratio from 1/5–1/10, linked to the increased amount of available acid reacting with the material [15,45].

The leaching percentage of all metals in the M2 stream remained constant over time (Fig. 4c), with values lower than 1.8 % for Y, 0.2 % for La, and higher than 98 % for Ni and Co. Leaching of REEs by H_3PO_4 is inverse of the temperature (Fig. 4d). Y showed the largest reduction (from 12.9 % at 25 °C to 1.8 % at 80 °C), while Nd was not leached under any of the tested conditions. The leaching percentage of La decreased from 3.0 % (25 °C) to 0.1 % (80 °C), while Ce and Pr decreased from 1.9 % and 2.1 % (25 °C), respectively, to 0 % (80 °C). This behavior can be attributed to the decreased solubility of REEPO_4 compounds with the rise in temperature [15]. The leaching efficiency of Ni increased from 93.5 % (25 °C) to 98.0 % (80 °C), while Co remained in the range of 98 % – 99 %. The optimal leaching parameters for the M2 stream with H_3PO_4 were defined as: acid concentration of 4.0 mol/L, temperature of 80 °C, S/L ratio of 1/10, and a reaction time of 30 min.

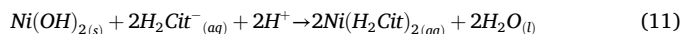
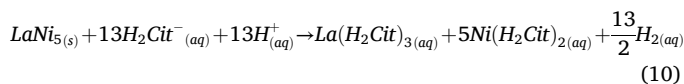
3.3.3. $\text{C}_6\text{H}_8\text{O}_7$ leaching

Leaching with organic acids generally occurs through chelation, protonation, and ligand exchange reactions that are closely related to the acid dissociation constant (pK_a) [46]. $\text{C}_6\text{H}_8\text{O}_7$ contains three carboxyl groups with dissociation constants $\text{pK}_{a1} = 3.13$, $\text{pK}_{a2} = 4.76$, and $\text{pK}_{a3} = 6.40$ at 25 °C (Eqs. 7–9; Figure S16) [47,48]. In this context, acids with higher pK_a values (such as $\text{C}_6\text{H}_8\text{O}_7$) dissociate gradually, releasing protons for leaching at a slower rate compared to acids with lower pK_a values (such as H_2SO_4 , $\text{pK}_{a1} = -3.0$ and $\text{pK}_{a2} = 1.99$). However, organic acids leaching is not solely dependent on the availability of protons provided by the acid but is also influenced by the nature of other ions in solution and the stability of the metal complexes

formed. The more stable the complexes, the higher the leaching efficiency [46].



The influence of acid concentration, temperature, S/L ratio, and reaction time on the leaching of metals from the M2 stream in $\text{C}_6\text{H}_8\text{O}_7$ is shown in Fig. 5. The stoichiometric acid ratio was calculated based on the reactions of LaNi_5 alloy and $\text{Ni}(\text{OH})_2$ with $\text{C}_6\text{H}_8\text{O}_7$ (Eqs. 10–11), assuming partial acid dissociation (Figure S16; Table S3), since within the pH range of the experiments, citric acid is predominantly present in the H_2Cit^- .



The leaching percentages remained constant, suggesting that neither acid concentration nor solution pH influenced the leaching process (Fig. 5a). This trend suggests that, under the tested conditions, the effective concentration of dissociated acid species rather than the total acid concentration controls the leaching efficiency. The pH of the leach solutions (Table S3) did not vary significantly with increasing acid concentration, evidencing the limited dissociation of the organic acid. This interpretation is consistent with the dissociation diagram of citric acid at 25 °C (Figure S16) [49], which shows that, at low pH, most acid species remain in protonated forms, reducing the availability of free citrate ions to complex or react with metal cations. Since the leaching percentages remained unchanged across the available acid concentrations, the lowest concentration (1.0 mol/L) was selected for subsequent experiments. Under this condition, leaching efficiencies were limited,

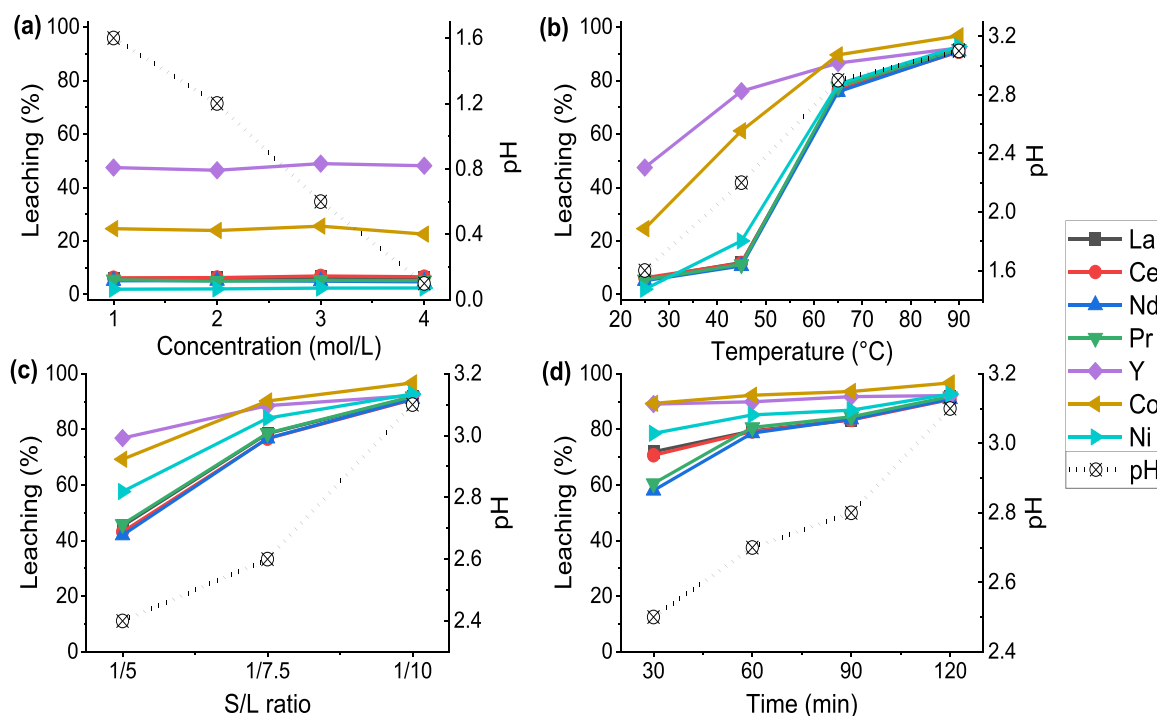


Figure 5. Leaching efficiency of REEs, Ni and Co with $C_6H_8O_7$ (a) by varying concentration (1.0–4.0 mol/L), at 25 °C, S/L ratio of 1/10 and 120 min; (b) by varying temperature (25–90 °C), acid concentration of 1.0 mol/L, S/L ratio of 1/10 and 120 min; (c) by varying S/L ratio (1/5–1/10), at 25 °C, acid concentration of 1.0 mol/L for 120 min; (d) by varying leaching time (30–120 min), at 25 °C, acid concentration of 1.0 mol/L and S/L ratio of 1/10.

with only 5.1 % – 6.2 % for LREEs, 47.4 % for Y, 2.0 % for Ni, and 25.0 % for Co.

Leaching increased proportionally with temperature (Fig. 5b). At 90 °C, more than 90 % of all elements were leached. Similar behavior was reported by Siyu et al. [27] in the leaching of Li-ion batteries for Ni, Co, and Mn. The protonation reactions of $C_6H_8O_7$ in aqueous media are endothermic and thermodynamically; therefore, higher temperatures favor both acid dissociation and the release of protons, increasing the concentration of reactive H^+ species. This enhances the proton-promoted dissolution of metallic phases. Additionally, temperature may accelerate ligand exchange and complexation reactions between citrate ions and metal cations, further improving metal solubility. Consequently, the overall leaching efficiency rises with temperature due to the combined effect of greater acid dissociation and faster reaction kinetics [11].

An increase in the leaching percentage of all metals was observed as the S/L ratio varied from 1/5–1/10 (Fig. 5c) due to the higher amount of acid available (0.5, 0.6 and 0.7 stoichiometric acid ratio – TableS3) and greater availability of protons in solution; low S/L ratios (as 1/7.5 and 1.10) result in a higher proportion of dissociated acid providing more ions available to react with metals [46].

An increase in the leaching percentage was observed over time (Fig. 5d), contrasting with previous experiments. Similar results were described by Siyu et al. [27] in leaching experiments of lithium-ion (NMC) batteries conducted from 30 to 100 min. The relation between increased leach efficiency and reaction time can be attributed to the metal dissolution in organic acids that occur progressively as ion dissociation takes place. Longer time reaction (than inorganic acids) enhances the availability of dissociated ions, leading to higher metal leaching efficiencies at 120 min [46].

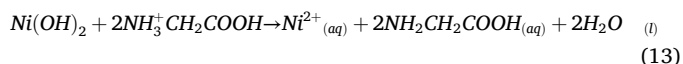
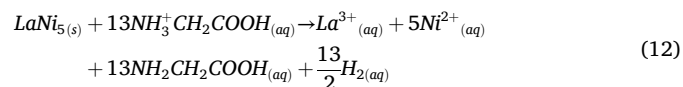
3.3.4. $C_2H_5NO_2$ leaching

$C_2H_5NO_2$ consists of an amino group (NH_2), a carboxyl group ($COOH$), and a central carbon bonded to two hydrogen atoms, and is considered one of the simplest amino acids known and is metabolized in most living organisms [50]. Depending on the pH of the solution

(Figure S17), $C_2H_5NO_2$ can exist in three forms: as a cation ($^+H_3NCH_2COOH$), as an anion ($H_2NCH_2COO^-$), and as a zwitterion ($^+H_3NCH_2COO^-$). In acidic pH ($pH < 2.35$), the cationic form predominates; between pH 2.35 and 9.78, the zwitterionic form is dominant; and in basic pH ($pH > 9.78$), the anionic form prevails [51,52].

No articles were found addressing the leaching of NiMH batteries with $C_2H_5NO_2$, and it was selected for testing due to both its novelty and environmental considerations [53]. However, studies have been reported on Li-ion batteries [30,54] and Ni-Cd batteries recycling [31,55]. $C_2H_5NO_2$ alone is not sufficient to achieve efficient metal leaching, and catalysts, pH modifiers, and oxidizing/reducing agents are often used in combination to enhance leaching percentages [53].

HNO_3 was added (2 mL per 50 mL of 2.0 mol/L glycine solution) to adjust the pH and compare the leaching efficiency (Fig. 6a). In the experiment conducted with $C_2H_5NO_2$ without the addition of HNO_3 (pH 6.3), the leaching percentages were below 4.0 %. As HNO_3 was added to adjust the pH (2.0), the leaching efficiency increased, reaching 5.7 % – 7.2 % for LREEs, 55.0 % for Y, 35.4 % for Co, and 5.3 % for Ni. The stoichiometric acid ratio was calculated (Eqs. 12–13), assuming all the acid was in its cationic form, leading to the release of La^{3+} and Ni^{2+} ions, as predicted by the thermodynamic simulation using Hydra-Medusa software (Figure S13). The results indicated that the process operated under sub-stoichiometric conditions, with a stoichiometric acid ratio of 0.8.



There is a positive effect of adding HNO_3 to the 2.0 mol/L $C_2H_5NO_2$ solution. Increasing the temperature to 90 °C enhanced leaching efficiencies reaching 83.8 % for Ni, 95.4 % for Co, 84.6 % for Y, and over 96.0 % for REEs (Fig. 6a). This fact could be attributed to increased

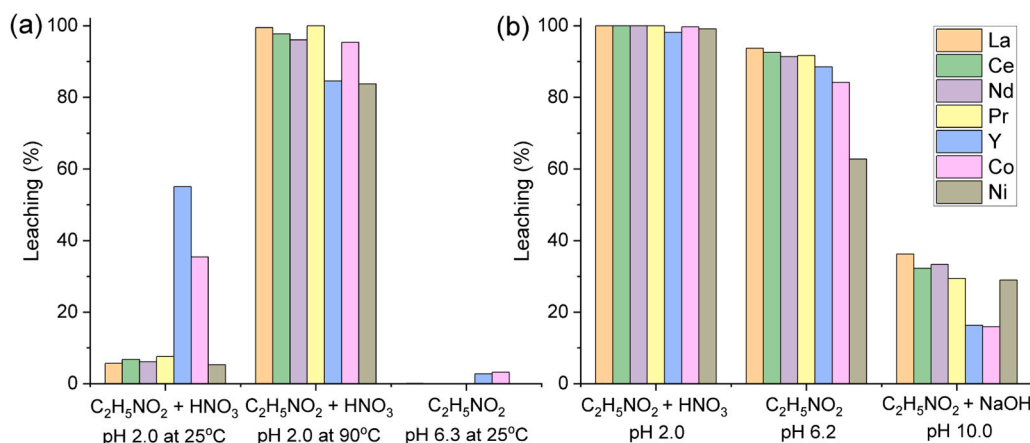


Fig. 6. Leaching efficiency of REEs, Ni and Co by $C_2H_5NO_2$ (a) with and without addition of HNO_3 , at 25 and 90 °C, S/L ratio of 1/10 and 120 min; (b) with and without addition of HNO_3 and NaOH, at 90 °C, S/L ratio of 1/20 and 120 min.

diffusivity, collision frequency, and mass transfer coefficient between the acid and the solid material [30,54,56]. The resulting leachate exhibited high viscosity, preventing proper filtration (S/L 1/10). Vacuum filtration required about four hours to complete. The causes of this behavior are beyond the scope of this work. To address this issue, a S/L ratio of 1/20 was tested, maintaining 2.0 mol/L $C_2H_5NO_2$, 90 °C, and 120 min. Additionally, we also tested the effect of pH (2.0, 6.3, and 10.0, Fig. 6b). The addition of NaOH to reach pH 10.0 was performed to allow a comparison of the leaching percentages of metals from the M2 stream with glycine in its three forms (Figure S17).

At 90 °C and an S/L ratio of 1/20 (pH 6.3 – without pH adjustment), the leaching efficiencies ranged from 88.5 % to 93.7 % for REEs, 62.8 % for Ni, and 84.2 % for Co. In contrast, it was observed to have the lowest leaching percentages at pH 10.0 (29.4 %–36.2 % for LREEs, 16.3 % for Y, 15.9 % for Co, and 29.0 % for Ni) probably attributed to the high concentration of hydroxide ions in alkali media, which compete with $C_2H_5NO_2$ for the coordination of metal ions and promote the formation of insoluble complexes (Figure S13) [55]. On the other hand, at pH 2.0, LREEs were fully leached (100 %), while Y, Co, and Ni achieved leaching efficiencies of 98.2 %, 99.7 %, and 99.2 %, respectively.

3.3.5. Comparison of leaching efficiency between the evaluated acids

The goal in this section was to identify the most feasible leaching agent for NiMH battery recycling, considering leaching efficiency, reagent cost, and scalability potential. The leaching experiments of the M2 stream with H_2SO_4 , H_3PO_4 , $C_6H_8O_7$, and $C_2H_5NO_2$ under varying parameters allowed the determination of the optimal operating conditions.

Table 4 summarizes the metal leaching efficiencies achieved under these optimal conditions for each acid, along with the estimated cost of the leaching stage. The cost estimation was performed by calculating the amount of acid required to leach 1 kg of black mass under the optimal acid concentration and S/L ratio defined for each system. The total acid volume was then multiplied by the corresponding market price per unit (sourced from “laballey.com,” accessed on April 14, 2025). This approach provides a preliminary estimation of the reagent cost associated with the leaching step, without accounting for additional operational factors such as energy consumption, equipment, or separation processes.

Leaching cost for H_2SO_4 was the lowest among all acids evaluated (\$53.9 per kilogram of black mass processed), and efficiencies of 100 % for the REEs, 97.7 % for Ni, and 99.2 % for Co were achieved. S/L ratio (1/7.5) and temperature (25 °C) used were the lowest compared to the other acids, while the reaction time (60 min) was longer than that required for H_3PO_4 (30 min) and shorter than for $C_6H_8O_7$ and $C_2H_5NO_2$ (120 min).

H_3PO_4 leaching cost was estimated at \$133.6, approximately 2.5 times higher than the H_2SO_4 leaching due to the higher concentration and S/L ratio required. To achieve a more precise and realistic comparison, a comprehensive cost analysis—including subsequent steps of purification and recovery of the target metals—should be conducted for both H_2SO_4 and H_3PO_4 processes to determine which route is more economically feasible. In our case, experiments with H_3PO_4 allowed selective leaching of Ni (98.0 %) and Co (98.9 %) while the REEs precipitated as phosphate. The highest acid concentration (4.0 mol/L)

Table 4

Leaching percentage (%) of metals in the M2 stream under the optimal operating conditions for each acid.

Acid	Conditions	La	Ce	Nd	Pr	Y	Al	Co	Fe	K	Mn	Ni	Zn	Reagent cost (US\$/kg of black mass)
H_2SO_4	2.0 mol/L 25 °C 60 min 1/7.5 S/L ratio	100	100	100	100	99.9	99.2	99.2	92.6	99.5	99.2	97.7	98.6	53.9
H_3PO_4	4.0 mol/L 80 °C 30 min 1/10 S/L ratio	0.1	-	-	-	1.8	96.8	98.9	97.5	100	91.6	98.0	94.2	133.6
$C_6H_8O_7$	1.0 mol/L 90 °C 120 min 1/10 S/L ratio	91.6	90.7	90.9	91.8	92.2	91.4	96.7	98.0	100	89.5	92.7	96.3	150.0
$C_2H_5NO_2$	2.0 mol/L 90 °C 120 min 1/20 S/L ratio pH 2.0 (HNO_3)	100	100	100	100	98.2	97.8	99.7	83.4	100	100	99.2	100	1192.1

and the shortest reaction time (30 min), but a temperature of 80 °C was necessary to enhance leaching selectivity. The resulting liquor would be ready for purification steps aimed at separating Ni and Co, such as precipitation and solvent extraction [2,57,58]. However, the REE phosphates formed are not commercially valuable. Further purification steps are needed, as the precipitate still contains other metals and remnants of separator material. A second leaching stage using HCl could be applied to dissolve the REE phosphates, followed by precipitation with oxalic acid [14].

The leaching cost with organic acids is higher compared with inorganic acids. Specifically, leaching with $C_6H_8O_7$ and $C_2H_5NO_2$ is 2.8 and 22.1 times more expensive, respectively, than with H_2SO_4 , making cost a key consideration in the selection of leaching agents for scale-up. However, despite their higher cost, the study of organic acids remains relevant due to their potential environmental advantages, such as biodegradability, lower toxicity, and reduced corrosiveness. These characteristics may simplify downstream processing, reduce equipment degradation, and contribute to the development of more sustainable and safer hydrometallurgical routes, factors that are increasingly valued in the context of green chemistry and circular economy strategies [22].

Leaching with organic acids required higher temperatures (90 °C) and longer reaction times (120 min) because the dissociation constants are higher than those of inorganic acids, and such conditions are needed to promote acid dissociation and enhance metal solubilization. Although the leaching efficiencies for Ni (92.7 %) and Co (96.7 %) were the lowest using $C_6H_8O_7$, it required the lowest acid concentration (1.0 mol/L) without effect on increasing concentration.

The use of $C_2H_5NO_2$ at pH 2.0 allows leaching percentages of 100 % for the LREEs, 98.2 % for Y, 99.7 % for Co, and 99.2 % for Ni. However, the main drawback of this approach is the use of the highest S/L ratio (1/20), as a lower S/L ratio (1/10) results in a highly viscous leachate. This poses potential scalability challenges for industrial applications, due to the need for larger reactor volumes and increased reagent consumption. To gain a better understanding of the leaching mechanism of this acid and to define more suitable conditions for potential scale-up, further studies should explore the effects of pH, $C_2H_5NO_2$ concentration, pH modifiers, and reaction time.

Based on the results, H_2SO_4 is considered here the most suitable leaching agent for NiMH black mass, offering the best combination of metal leaching efficiency, mild operating conditions, and the lowest reaction cost (Table 4). This does not imply that leaching with other acids is not feasible. Organic acids, for example, may offer benefits in terms of safety and biodegradability, while H_3PO_4 demonstrates potential advantages in selectivity.

Regarding environmental concerns, the use of inorganic acids can produce highly acidic effluents that require careful neutralization and proper disposal to prevent environmental contamination. In contrast, the use of organic acids can reduce the generation of hazardous waste and enable the development of more sustainable recycling processes. Nevertheless, even organic acids may pose risks at high concentrations or volumes, and effluent management, recycling, or recovery strategies should be implemented to minimize environmental impact. A comprehensive assessment of the environmental implications of each leaching agent, including factors such as toxicity, biodegradability, and waste treatment requirements, will be necessary to support a well-informed and robust decision regarding the most sustainable choice of reagent.

4. Conclusion

Approximately 8.6 kg of Ni, 2.0 kg of REEs, and 0.5 kg of Co could potentially recover from a 37.2 kg HEV NiMH battery pack. By comminution and physical separation, it was possible to separate approximately 97 % of the REEs, 72 % of the Ni, and 84 % of the Co in a single product stream (M2) for leaching experiments.

This work enabled a direct comparison of the leaching efficiencies of NiMH batteries using organic and inorganic acids, and the identification

of operational parameters that maximize metal recovery while considering cost and feasibility. The leaching results confirmed the feasibility of using organic acids ($C_6H_8O_7$ and $C_2H_5NO_2$), achieving efficiencies comparable to inorganic acids (H_2SO_4 and H_3PO_4). However, organic acid-based processes require higher temperatures and longer reaction times to compensate for their higher dissociation constants and slower proton release kinetics. These factors, combined with the typically higher cost of organic acids, contribute to increased operational expenses for the leaching stage. Among the tested reagents, $C_2H_5NO_2$ presented the greatest challenges in terms of scalability due to the high S/L ratio (1/20), but this study demonstrates for the first time its feasibility as a leaching agent for NiMH battery black mass.

Temperature proved to be a key parameter in leaching with H_3PO_4 , as it reduces the solubility of REE phosphates while enhancing the leaching of Ni and Co, enabling their selective separation. Although effective, this strategy required a high acid concentration (4.0 mol/L – stoichiometric ratio of 3.9) and the resulting insoluble REE phosphates necessitate additional recovery steps before market. Nevertheless, this route is particularly attractive when the primary objective is the recovery of transition metals, as it allows the removal of REEs in a single step, streamlining the downstream processing of Ni and Co.

The best overall performance was obtained with H_2SO_4 . It combines leaching efficiencies above 97 % for the critical metals with the lowest operational cost, making it particularly suitable for large-scale applications. Beyond its effectiveness, H_2SO_4 is widely available, cost-efficient, and compatible with standard hydrometallurgical infrastructure, which facilitates its integration into existing industrial processes. Taken together, these factors make H_2SO_4 the most practical and scalable option for the recovery of valuable metals from NiMH battery black mass, reinforcing its role as the preferred leaching agent in hydrometallurgical processing.

Future studies should focus on optimizing processes based on organic acids to improve scalability, reduce reagent consumption, and enhance overall process efficiency. In addition, a comprehensive techno-economic and environmental assessment of the full recycling process—including leaching, separation, and purification steps—will be essential to support the industrial implementation of these methods and to ensure that the selected leaching routes are both economically viable and environmentally sustainable.

CRedit authorship contribution statement

Denise Croce Romano Espinosa: Supervision, Resources, Project administration, Funding acquisition, Conceptualization. **Amilton Barbosa Botelho Junior:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Project administration, Methodology, Investigation, Conceptualization. **Daniel Sánchez Piloto:** Writing – review & editing, Writing – original draft, Methodology, Investigation, Formal analysis, Data curation, Conceptualization.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

The authors thank the University of Sao Paulo for supporting this project and Fundação de Amparo à Pesquisa do Estado de São Paulo and Capes (2019/11866–5, 2021/14842–0, and 2023/01032–5) for the financial support. ABBJR would like to thank MIT Climate Grand Challenges and the Department of Chemical Engineering at NTNU for the financial support.

Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at [doi:10.1016/j.jece.2026.121066](https://doi.org/10.1016/j.jece.2026.121066).

Data Availability

Data will be made available on request.

References

- [1] D. Kolodyńska, K. Burdzy, S. Hunger, A. Aurich, Y. Ju, Green extractants in assisting recovery of REEs: a case study, *Molecules* 28 (2023), <https://doi.org/10.3390/molecules28030965>.
- [2] M. Takano, S. Asano, M. Goto, Recovery of nickel, cobalt and rare-earth elements from spent nickel-metal-hydride battery: laboratory tests and pilot trials, *Hydrometallurgy* 209 (2022), <https://doi.org/10.1016/j.hydromet.2022.105826>.
- [3] Coherent Market Insights, Nickel metal Hydride battery market size and share analysis-growth trends and forecasts (2023-2030), Coherent Market Insights, 2024. (<https://www.coherentmarketinsights.com/industry-reports/nickel-metal-hydride-battery-market/>) (accessed August 19, 2024).
- [4] Verified Market Research, Nickel Metal Hydride (Ni-MH) Battery Market Size And Forecast, Nickel Metal Hydride (Ni-MH) Battery Market Size And Forecast, 2024. (<https://www.verifiedmarketresearch.com/product/nickel-metal-hydride-ni-mh-battery-market/>) (accessed December 3, 2024).
- [5] Markets and Markets, Nickel Metal Hydride (NiMH) Battery Market, Markets and Markets, 2023. (<https://www.marketsandmarkets.com/Market-Reports/nickel-metal-hydride-nimh-battery-market-128735977.html>) (accessed August 19, 2024).
- [6] Statista, statistic_id1339880_battery-market-size-worldwide-by-technology-2018-2030, Size of the Global Battery Market from 2018 to 2021, with a Forecast through 2030, by Technology (in Million U.S. Dollars), 2024.
- [7] M. Fetcenko, J. Koch, M. Zelinsky, Nickel-metal hydride and nickel-zinc batteries for hybrid electric vehicles and battery electric vehicles. *Advances in Battery Technologies for Electric Vehicles*, Elsevier, 2015, pp. 103–126, <https://doi.org/10.1016/B978-1-78242-377-5.00006-6>.
- [8] E.H. Tanabe, D.F. Schlemmer, M.L. Aguiar, G.L. Dotto, D.A. Bertuol, Recovery of valuable materials from spent NiMH batteries using spouted bed elutriation, *J. Environ. Manag.* 171 (2016) 177–183, <https://doi.org/10.1016/j.jenvman.2016.02.011>.
- [9] S. Al-Thyabat, T. Nakamura, E. Shibata, A. Iizuka, Adaptation of minerals processing operations for lithium-ion (LiBs) and nickel metal hydride (NiMH) batteries recycling: Critical review, *Min. Eng.* 45 (2013) 4–17, <https://doi.org/10.1016/j.mineng.2012.12.005>.
- [10] S. Pradhan, R. Nayak, S. Mishra, A review on the recovery of metal values from spent nickel metal hydride and lithium-ion batteries, *Int. J. Environ. Sci. Technol.* 19 (2022) 4537–4554, <https://doi.org/10.1007/s13762-021-03356-5>.
- [11] H. Zhi, S. Ni, X. Su, W. Xie, H. Zhang, X. Sun, Separation and recovery of rare earth from waste nickel-metal hydride batteries by phosphate based extraction-precipitation, *J. Rare Earths* 40 (2022) 974–980, <https://doi.org/10.1016/j.jre.2021.04.012>.
- [12] S. Mir, N. Shukla, N. Dhawan, Investigation of microwave and thermal processing of electrode material of end-of-life Ni-MH battery, *JOM* 73 (2021) 951–961, <https://doi.org/10.1007/s11837-020-04530-9>.
- [13] M. Petranikova, I. Herdizik-Koniecko, B.M. Steenari, C. Ekberg, Hydrometallurgical processes for recovery of valuable and critical metals from spent car NiMH batteries optimized in a pilot plant scale, *Hydrometallurgy* 171 (2017) 128–141, <https://doi.org/10.1016/j.hydromet.2017.05.006>.
- [14] J. Lie, J.C. Liu, Selective recovery of rare earth elements (REEs) from spent NiMH batteries by two-stage acid leaching, *J. Environ. Chem. Eng.* 9 (2021), <https://doi.org/10.1016/j.jece.2021.106084>.
- [15] K. Tanji, K. Ouzauit, M. Belghiti, I. Lamsayety, H. Faqir, I. Benzakour, Hydrometallurgical two stage process for preparation of (Nd, La, Ce)2O3 from end-of-life NiMH batteries, *J. Rare Earths* 42 (2024) 415–423, <https://doi.org/10.1016/j.jre.2023.02.020>.
- [16] L. Cassayre, B. Guzhev, M. Zielinski, B. Biscans, Chemical processes for the recovery of valuable metals from spent nickel metal hydride batteries: a review, *Renew. Sustain. Energy Rev.* 170 (2022), <https://doi.org/10.1016/j.rser.2022.112983>.
- [17] P.R. Gismonti, J.F. Paulino, J.C. Afonso, Recovery of metals from electroactive components of spent ni-mh batteries after leaching with formic acid, *Detritus* 14 (2021) 68–77, <https://doi.org/10.31025/2611-4135/2021.14063>.
- [18] J. Constantine, J. Lie, J.C. Liu, Recovery of rare earth elements from spent NiMH batteries using subcritical water extraction with citric acid, *J. Environ. Chem. Eng.* 10 (2022), <https://doi.org/10.1016/j.jece.2022.108000>.
- [19] A.Z. Colmenares, J.D. Salaverría, P. Delvasto, Characterization of the chemical compounds obtained after using acetic acid as leaching agent in the hydrometallurgical treatment of spent Ni-MH batteries, *Prod. Y. Limpia* 13 (2018) 19–29, <https://doi.org/10.22507/pml.v13n1a2>.
- [20] A.A.L. Marins, S.G. Banhos, E.J.B. Muri, R.V. Rodrigues, P.C.M. Cruz, M.B.J. G. Freitas, Synthesis by coprecipitation with oxalic acid of rare earth and nickel oxides from the anode of spent Ni-MH batteries and its electrochemical properties, *Mater. Chem. Phys.* 242 (2020), <https://doi.org/10.1016/j.matchemphys.2019.122440>.
- [21] S. Singh, P. Srivastava, S. Deep, Ashish, P. Shukla, H. Rizvi, R. Sinha, Fermentation Strategies for Organic Acid Production, in: P. Verma (Ed.), *Industrial Microbiology and Biotechnology*, Springer Singapore, Singapore, 2022, pp. 379–425, https://doi.org/10.1007/978-981-16-5214-1_14.
- [22] E. Gerold, C. Schinnerl, H. Antrekowitsch, Critical evaluation of the potential of organic acids for the environmentally friendly recycling of spent lithium-ion batteries, *Recycling* 7 (2022), <https://doi.org/10.3390/recycling7010004>.
- [23] A.B. Botelho Junior, Sustainable mining – unlocking resources towards circular economy to meet energy transition through electrochemistry, *J. Environ. Chem. Eng.* 13 (2025) 116600, <https://doi.org/10.1016/j.jece.2025.116600>.
- [24] H. Salehi, S. Maroufi, S.S. Mofarah, R.K. Nekouei, V. Sahajwalla, Recovery of rare earth metals from Ni-MH batteries: a comprehensive review, *Renew. Sustain. Energy Rev.* 178 (2023), <https://doi.org/10.1016/j.rser.2023.113248>.
- [25] S. Mousavinezhad, S. Kadivar, E. Vahidi, Comparative life cycle analysis of critical materials recovery from spent Li-ion batteries, *J. Environ. Manag.* 339 (2023), <https://doi.org/10.1016/j.jenvman.2023.117887>.
- [26] O.E. Roshdy, E.A. Haggag, A.M. Masoud, M. Bertau, N. Haneklaus, S. Pavón, A.E. M. Hussein, Y.M. Khawassek, M.H. Taha, Leaching of rare earths from Abu Tartur (Egypt) phosphate rock with phosphoric acid, *J. Mater. Cycles Waste Manag.* 25 (2023) 501–517, <https://doi.org/10.1007/s10163-022-01558-8>.
- [27] G. Siyu, D. Enhua, L. Bingguo, Y. Chao, N. Yifan, J. Guangxiong, C. Wang, H. Keren, G. Shenghui, Z. Libo, Eco-friendly closed-loop recycling of nickel, cobalt, manganese, and lithium from spent ternary lithium-ion battery cathodes, *Sep. Purif. Technol.* 348 (2024), <https://doi.org/10.1016/j.seppur.2024.127771>.
- [28] J. Wang, K. Huang, H. Dong, Y. Lu, K. Liu, Z. Chen, X. Shan, G. Huang, L. Wei, A green process for recycling and synthesis of cathode materials LiMn2O4 from spent lithium-ion batteries using citric acid, *RSC Adv.* 12 (2022) 23683–23691, <https://doi.org/10.1039/D2RA04391B>.
- [29] Y. Zhang, Y. Wang, Z. Han, M. Jin, X. Li, Y. Li, X. Kong, Z. Zhang, X. Zhang, Kinetic optimization in green metal recovery: Citric acid vs glycine for sustainable Li-ion battery recycling, *Ecotoxicol. Environ. Saf.* 302 (2025), <https://doi.org/10.1016/j.ecoenv.2025.118746>.
- [30] J. Wu, J. Ahn, J. Lee, A sustainable complexation leaching of critical metals from spent lithium-ion batteries by glycine in a neutral solution, *Min. Met. Explor* 41 (2024) 1605–1617, <https://doi.org/10.1007/s42461-024-01040-z>.
- [31] M. Nazaralilou, D.F. Haghsheenas, S. Firoozi, Selective leaching of Ni from Cd-depleted spent Ni-Cd batteries by glycine-citrate solution, *Min. Eng.* 203 (2023), <https://doi.org/10.1016/j.mineng.2023.108332>.
- [32] A.B. Botelho Junior, D.V. da Silva, A.M.N. de O. Lima, R.P. de Oliveira, L.A. Gobo, E.A. Kumoto, A. Ferrarese, J.A.S. Tenório, D.C.R. Espinosa, LAREX-Tupy Process: Recycling of Li-Ion Batteries from Electric Vehicles by Hydrometallurgical Route Towards Circular Economy, in: TMS 2024 153rd Annual Meeting & Exhibition Supplemental Proceedings, Springer Nature Switzerland, Cham, 2024, pp. 1920–1927.
- [33] R.H. de Castro, D.C. Romano Espinosa, L.A. Gobo, E.A. Kumoto, A.B. Botelho Junior, J.A.S. Tenorio, Design of recycling processes for NCA-Type Li-ion batteries from electric vehicles toward the circular economy, *Energy Fuels* 38 (2024) 5545–5557, <https://doi.org/10.1021/acs.energyfuels.3c04904>.
- [34] D. Romero Guillén, J. Guimarães Sanches, A. Barbosa Botelho Junior, L.A. Gobo, M. Guimarães Bergerman, D.C. Romano Espinosa, J.A.S. Tenório, Physical process for Li-ion battery recycling from electric vehicles, *Ind. Eng. Chem. Res.* (2024), <https://doi.org/10.1021/acs.iecr.4c03271>.
- [35] Kivanc. Korkmaz, Recovery of Rare Earth Elements from spent Nickel-Metal Hydride Batteries from Hybrid Electric Vehicles, 2021.
- [36] K. Larsson, C. Ekberg, A. Ødegaard-Jensen, Dissolution and characterization of HEV NiMH batteries, *Waste Manag.* 33 (2013) 689–698, <https://doi.org/10.1016/j.wasman.2012.06.001>.
- [37] M.L. Castro, D. María, G. Montes, O. Yemha, J. Iván, A. González, Recuperación de metales a partir de baterías níquel-hidruro metálico usando disolventes eutécticos profundos, *Exploraciones, Intercamb. Y. Relac. Entre El Diseño Y. La Tecnol. ía* (2018) 57–79. (<http://zaloamati.azc.uam.mx/handle/11191/6806>).
- [38] D.M. Navarrete, M.R. Esquivel, M.H. Rodriguez, Extracción de níquel a partir de baterías agotadas de ni-mh por vía hidrometalúrgica, 2015, 8.
- [39] K. Korkmaz, M. Alemrajabi, Å.C. Rasmuson, K.M. Forsberg, Sustainable hydrometallurgical recovery of valuable elements from spent nickel-metal hydride HEV batteries, *Metals* 8 (2018), <https://doi.org/10.3390/met8121062>.
- [40] L. Ouyang, J. Huang, H. Wang, J. Liu, M. Zhu, Progress of hydrogen storage alloys for Ni-MH rechargeable power batteries in electric vehicles: a review, *Mater. Chem. Phys.* 200 (2017) 164–178, <https://doi.org/10.1016/j.matchemphys.2017.07.002>.
- [41] Shanghai Metals Market, Shanghai Metals Market, Rare Earth Oxides Prices, 2024. (<https://www.metal.com/>) (accessed September 26, 2024).
- [42] C. Tunstu, M. Petranikova, M. Gergorić, C. Ekberg, T. Retegan, Reclaiming rare earth elements from end-of-life products: a review of the perspectives for urban mining using hydrometallurgical unit operations, *Hydrometallurgy* 156 (2015) 239–258, <https://doi.org/10.1016/j.hydromet.2015.06.007>.
- [43] M.S. Gasser, M.I. Aly, Separation and recovery of rare earth elements from spent nickel-metal-hydride batteries using synthetic adsorbent, *Int. J. Min. Process* 121 (2013) 31–38, <https://doi.org/10.1016/j.minpro.2013.02.012>.
- [44] A. Sobianowska-Turek, Hydrometallurgical recovery of metals: Ce, La, Co, Fe, Mn, Ni and Zn from the stream of used Ni-MH cells, *Waste Manag.* 77 (2018) 213–219, <https://doi.org/10.1016/j.wasman.2018.03.046>.
- [45] S. xi WU, L. sheng ZHAO, L. shi WANG, X. wei HUANG, J. shi DONG, Z. yu FENG, D. li CUI, L. feng ZHANG, Dissolution behaviors of rare earth elements in

- phosphoric acid solutions, *Trans. Nonferrous Met. Soc. China (Engl. Ed.)* 28 (2018) 2375–2382, [https://doi.org/10.1016/S1003-6326\(18\)64883-6](https://doi.org/10.1016/S1003-6326(18)64883-6).
- [46] S. Belfqueh, A. Seron, S. Chapron, Evaluating organic acids as alternative leaching reagents for rare earth elements recovery from NdFeB magnets, 2022.
- [47] W. Astuti, T. Hirajima, K. Sasaki, N. Okibe, Comparison of atmospheric citric acid leaching kinetics of nickel from different Indonesian saprolitic ores, *Hydrometallurgy* 161 (2016) 138–151, <https://doi.org/10.1016/j.hydromet.2015.12.015>.
- [48] M. Gergoric, C. Ravau, B.M. Steenari, F. Espegren, T. Retegan, Leaching and recovery of rare-earth elements from neodymium magnet waste using organic acids, *Metals* 8 (2018), <https://doi.org/10.3390/met8090721>.
- [49] C. Payan, A.L. Gancel, M. Jourdes, M. Christmann, P.L. Teissedre, Wine acidification methods: a review, *Oeno One* 57 (2023) 113–126, <https://doi.org/10.20870/oeno-one.2023.57.3.7476>.
- [50] H. Fathollahzadeh, H.N. Khaleque, J. Eksteen, A.H. Kaksonen, E.L.J. Watkin, Effect of glycine on bioleaching of rare earth elements from Western Australian monazite by heterotrophic and autotrophic microorganisms, *Hydrometallurgy* 189 (2019), <https://doi.org/10.1016/j.hydromet.2019.105137>.
- [51] C.P. Broeksma, C. Dorfling, Evaluating glycine as an alternative lixiviant for copper recovery from waste printed circuit boards, *Waste Manag.* 163 (2023) 96–107, <https://doi.org/10.1016/j.wasman.2023.03.030>.
- [52] E.A. Oraby, J.J. Eksteen, The selective leaching of copper from a gold-copper concentrate in glycine solutions, *Hydrometallurgy* 150 (2014) 14–19, <https://doi.org/10.1016/j.hydromet.2014.09.005>.
- [53] H. Li, Z. Deng, E. Oraby, J. Eksteen, Amino acids as lixiviants for metals extraction from natural and secondary resources with emphasis on glycine: a literature review, *Hydrometallurgy* 216 (2023), <https://doi.org/10.1016/j.hydromet.2022.106008>.
- [54] M. Chen, R. Wang, Y. Qi, Y. Han, R. Wang, J. Fu, F. Meng, X. Yi, J. Huang, J. Shu, Cobalt and lithium leaching from waste lithium ion batteries by glycine, *J. Power Sources* 482 (2021), <https://doi.org/10.1016/j.jpowsour.2020.228942>.
- [55] H. Oghabi, D.F. Haghshenas, S. Firoozi, Selective separation of Cd from spent Ni-Cd battery using glycine as an eco-friendly leachant and its recovery as CdS nanoparticles, *Sep Purif. Technol.* 242 (2020), <https://doi.org/10.1016/j.seppur.2020.116832>.
- [56] R. Rautela, B.R. Yadav, S. Kumar, Enhancing the efficiency of metals extraction from waste lithium-ion batteries through glycine leaching using response surface methodology for economic feasibility, *J. Clean. Prod.* 447 (2024), <https://doi.org/10.1016/j.jclepro.2024.141602>.
- [57] D.A. Bertuol, A.M. Bernardes, J.A.S. Tenório, Spent NiMH batteries-The role of selective precipitation in the recovery of valuable metals, *J. Power Sources* 193 (2009) 914–923, <https://doi.org/10.1016/j.jpowsour.2009.05.014>.
- [58] M. Petranikova, B. Ebin, C. Tunsu, Selective recovery of cobalt from the secondary streams after NiMH batteries processing using Cyanex 301, *Waste Manag.* 83 (2019) 194–201, <https://doi.org/10.1016/j.wasman.2018.11.022>.